Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Proposal

Proponent/Researcher:	**EW10**
Project Title: LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc.	
Introduction	Learning and work immersion are important parts of the Senior High School (SHS) program because they help students develop knowledge, skills, and real-world experience. However, many students still experience difficulties in studying their lessons using traditional methods, which can sometimes make learning boring and less engaging. At the same time, some schools still use manual methods such as paper in recording attendance and daily journals during work immersion, which can sometimes lead to lost records or difficulty in monitoring student activities. With the development of modern technology, schools can now use digital systems to improve both learning and student monitoring. Artificial Intelligence (AI) is now being used in educational systems to help students study more effectively by generating quizzes, activities, and reviewers from learning materials. Digital systems can also help teachers and immersion coordinators monitor students' attendance and activities more efficiently. Because of this, the researchers propose LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The system will allow teachers to upload learning materials such as PowerPoint and document files, which will then be analyzed by AI to generate quizzes, activities, and reviewers that students can use for studying. The system will also include a ranking or leaderboard feature to motivate students to improve their performance. In addition, the system will include a digital platform where students can record their Time-In and Time-Out during their work immersion and submit their daily journal entries. Teachers and immersion coordinators will be able to monitor student attendance, journals, and learning progress through a web-based dashboard. The main goal of the system is to make learning more interactive and engaging while also making the monitoring of work immersion activities easier, more organized, and more efficient for both students and teachers.
Statement of the Problem	This study aims to develop LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. Many students still rely on traditional study methods, which can sometimes make learning less interactive and engaging. In addition, some schools still use manual methods such as paper in

	recording attendance and daily journals during work immersion, which may lead to lost or incomplete records. Specifically, it seeks to answer the following questions: 1. How can LearnIQ Track be designed and developed to help students review their lessons using AI-generated quizzes, activities, and reviewers? 2. How can the system allow teachers to upload learning materials that can be analyzed by AI to generate learning content? 3. How can the system record students' Time-In and Time-Out during their work immersion? 4. How can the system allow students to submit their daily journals or reflections about their immersion activities? 5. How can the system help teachers and immersion coordinators monitor students' learning progress, attendance, and journal entries? 6. How can the system make learning and immersion monitoring more organized, interactive, and efficient?
Objective of the Study	The main objective of this study is to develop LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. that will help students review their lessons and record their work immersion activities using a digital platform. Specifically, this study aims to: 1. To design and develop an AI-based learning system that can generate quizzes, activities, and reviewers from uploaded learning materials. 2. To allow teachers to upload lesson materials such as PowerPoint and document files into the system. 3. To develop a leaderboard or ranking system that records student scores and performance. 4. To create a digital attendance system that records students' Time-In and Time-Out during work immersion. 5. To allow students to submit their daily journals or reflections about their immersion activities. 6. To provide a platform where teachers and immersion coordinators can monitor students' learning progress, attendance, and journal submissions.
Scope and Limitation of the Study	This study focuses on the development of LearnIQ Track, an AI-based learning game and digital work immersion monitoring system for Senior High School students of Paete Science and Business College Inc. The system will allow teachers to upload learning materials such as PowerPoint presentations and document files. The AI will analyze these materials and automatically generate quizzes, activities, and reviewers that students can use for studying. The system will also include features for recording students' Time-In and Time-Out during their work immersion and for submitting daily journal entries about their immersion activities. Teachers and immersion coordinators will be

	able to monitor student attendance, journal entries, and learning progress through a web-based dashboard. However, the system has some limitations. The AI-generated questions and reviewers will depend on the quality of the uploaded learning materials and may still need checking by teachers. The system also requires an internet connection to access its features. In addition, the system will only be used by Senior High School students and authorized personnel of PSBC Pagsanjan Branch and will only support selected file formats such as PowerPoint and document files.
Review of Literatures and Related Systems	The researchers collected the information that are greatly related to the study. These related studies and literature are compiled in this section as a basis in the development of the system. Related studies and projects are listed below in thematic order: **Theme 1: Artificial Intelligence in Education** Artificial Intelligence (AI) is becoming an important technology in education because it can help improve how students learn. AI systems can analyze learning materials and student performance to provide personalized learning experiences. According to Holmes et al. (2022), AI technologies can support teaching and learning by providing automated feedback, intelligent tutoring systems, and adaptive learning tools. These technologies help students understand lessons more effectively and allow teachers to manage learning activities more efficiently. AI can also assist in creating learning materials automatically. Zawacki-Richter et al. (2020) explained that AI applications in education can analyze data from learning systems to improve instructional design and support students' academic performance. **Theme 2: Automated Quiz Generation** Automated question generation is one of the useful applications of AI in educational technology. It allows systems to automatically create quizzes or test questions from learning materials. According to Kurdi et al. (2021), automatic question generation uses natural language processing techniques to produce questions from text-based materials. This helps reduce the workload of teachers and allows students to practice answering questions more frequently. Similarly, Liu et al. (2023) explained that deep learning techniques can be used to generate multiple-choice questions from educational texts. These systems can analyze lesson content and create relevant questions that help students review their knowledge. **Theme 3: Game-Based Learning and Student Engagement** Game-based learning is widely used to make learning more engaging and interactive for students. Educational systems that include game elements such as points, leaderboards, and rewards can motivate students to participate more actively in learning activities. According to Dichev and Dicheva (2020), gamification in education helps increase student motivation and engagement when game elements are integrated into learning platforms.

In addition, research by Al-Adwan et al. (2022) showed that gamified learning systems improve students' participation and help them develop better learning outcomes. These systems encourage students to study while enjoying the learning process.

Theme 4: Web-Based Learning Systems (Online Learning Platforms)

Web-based learning systems allow students to access educational materials anytime and anywhere using the internet. These systems support flexible learning and allow students to continue studying even outside the classroom. According to Dhawan (2020), online learning platforms became very important during emergencies such as school closures because they allowed students and teachers to continue education through digital tools.

Web-based platforms also support independent learning by giving students access to lessons, quizzes, and other educational resources. These systems help improve learning accessibility and provide opportunities for students to study at their own pace.

Theme 5: Work Immersion Programs in Senior High School

Work immersion programs are designed to help students gain real-world work experience and develop practical skills related to their field of study. These programs allow students to apply the knowledge they learned in school to actual workplace situations. According to Meriam M. Alawin-Ligwang (2025), work immersion helps improve students' technical skills, workplace behavior, and readiness for employment. The study found that students who participate in immersion programs become more confident in performing real work tasks and communicating in professional environments.

Similarly, Democrito C. Cabile Jr. (2024) evaluated the implementation of work immersion programs and found that proper supervision and monitoring of students during immersion help improve their performance and learning experience. The study emphasized that structured immersion activities allow students to gain meaningful workplace knowledge and improve their motivation to complete tasks effectively.

In another study, Jocelyn J. Pestijo (2023) explained that work immersion helps students develop responsibility, discipline, and communication skills in real workplace settings. The research also highlighted that proper monitoring and documentation of student activities during immersion are important to ensure that students complete their required training hours and learning objectives.

Theme 6: Digital Internship and Work Immersion Monitoring Systems

With the help of modern technology, many schools now use digital systems to monitor internship and work immersion activities. These systems allow students to record their attendance and submit reports online while teachers and supervisors can easily track their progress.

According to Kelfin Nurfaizi and Djarot Hindarto (2023), web-based internship attendance systems improve the accuracy of attendance records and reduce manual errors compared to traditional paper-based methods. The study

also explained that digital monitoring platforms help administrators manage student activities more efficiently.

Similarly, Rica A. Barrocan et al. (2025) developed a web-based internship monitoring system that allows supervisors to track student attendance and progress in real time. Their findings showed that digital monitoring systems help improve organization, transparency, and efficiency in internship programs.

Another study by Thanakorn Sangkudluo (2023) also emphasized that online monitoring platforms help educators easily supervise student internship activities and improve communication between students and supervisors.

Conceptual Framework

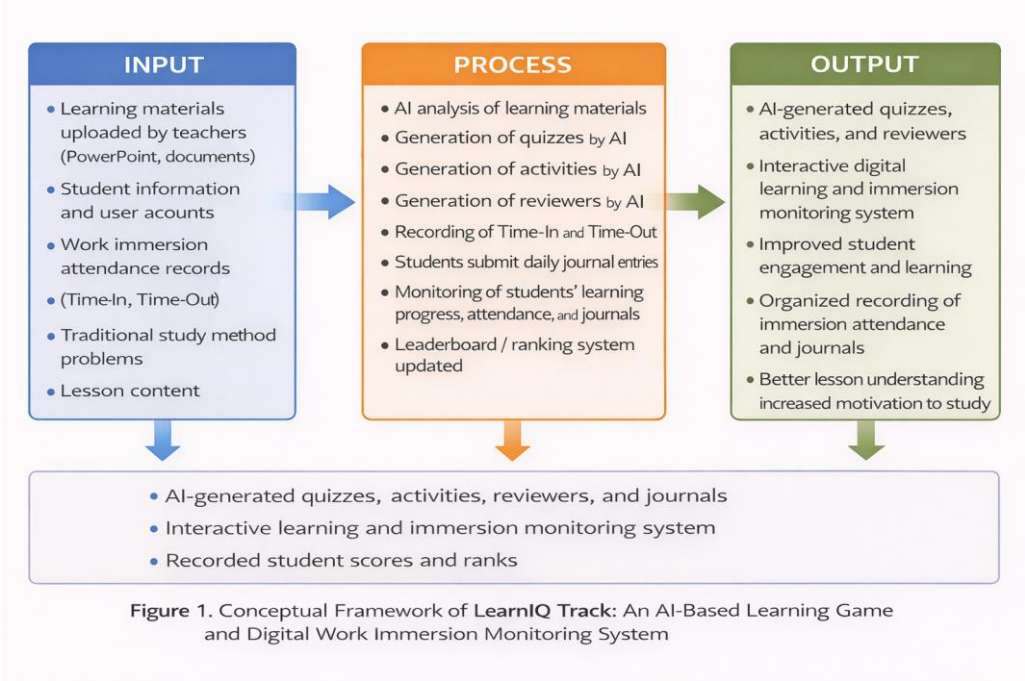

Figure 1. Conceptual Framework of LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc.

The conceptual framework of the study illustrates how the research will be conducted. The framework reflects the research objectives and activities to be conducted

SWOT ANALYSIS

Strengths	Weaknesses	Opportunities	Threats
• Uses Artificial Intelligence (AI) to generate quizzes, activities, and reviewers. • Helps students study in a	• The system depends on internet connection. • AI-generated content may still need checking by teachers.	• The system can be expanded and used by other schools. • Additional features such as reports and	• Technical issues such as system bugs or server downtime may affect the system. • Internet connectivit

	more interactive and engaging way. • Allows digital recording of work immersion attendance and journals. • Helps teachers and coordinators easily monitor students.	• Some users may not be familiar with digital systems.	analytics can be added in the future. • The system can help schools improve their digital learning methods.	y problems may limit system usage. • Some users may prefer traditional methods instead of using technology.

References

1. Holmes, W., Bialik, M., & Fadel, C. (2022). Artificial intelligence in education: Promise and implications for teaching and learning. *AI & Society*. https://doi.org/10.1007/s00146-021-01205-4
2. Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2020). Systematic review of research on artificial intelligence applications in higher education. *International Journal of Educational Technology in Higher Education, 17*(1). https://doi.org/10.1186/s41239-019-0171-0
3. Kurdi, G., Leo, J., Parsia, B., Sattler, U., & Al-Emari, S. (2021). A systematic review of automatic question generation for educational purposes. *International Journal of Artificial Intelligence in Education*. https://doi.org/10.1007/s40593-020-00224-0
4. Liu, M., Huang, S., & Zhang, Y. (2023). Deep learning for automatic question generation in educational systems. *Computers & Education: Artificial Intelligence, 4*. https://doi.org/10.1016/j.caeai.2022.100115
5. Dichev, C., & Dicheva, D. (2020). Gamifying education: What is known and what is next? *Educational Technology & Society, 23*(3), 75–88. https://www.jstor.org/stable/26981756
6. Al-Adwan, A. S., Al-Adwan, A., & Smedley, J. (2022). Exploring students' acceptance of e-learning systems in higher education. *Education and Information Technologies*. https://doi.org/10.1007/s10639-021-10761-7
7. Dhawan, S. (2020). Online learning: A panacea in the time of COVID-19 crisis. *Educational Technology Research and Development, 68*(5), 2385–2402. https://doi.org/10.1007/s11423-020-09806-z
8. Alawin-Ligwang, M. M. (2025). Work immersion performance of senior high school students: A foundation for enhancing implementation. *Journal of Education, Society and Behavioural Science*. https://doi.org/10.9734/jesbs/2025/v38i41403

9. Anajao, L. J. L., Intano, J. A., & Sumaoy, R. C. (2025). Industry feedback on work immersion: A qualitative exploration. *International Journal of Research and Innovation in Social Science*. https://rsisinternational.org/journals/ijriss/articles/industry-feedback-on-work-immersion-a-qualitative-exploration/

10. Barrocan, R. A., Calzo, M. J. R., Carambas, C. B., Tugade, M. J. E., & Bartolome, M. B. (2025). An architectural approach of a web-based monitoring system for efficient internship time tracking. *International Journal of Research and Innovation in Applied Science*. https://rsisinternational.org/journals/ijrias/articles/an-architectural-approach-of-a-web-based-monitoring-system-for-efficient-internship-time-tracking/

11. Cabile Jr., D. C. (2024). Implementation of work immersion in the new normal: Basis for students' quality performance and satisfaction. *International Journal of Multidisciplinary: Applied Business and Education Research*. https://www.babmrjournal.org/index.php/ijmaber/article/view/1587

12. Nurfaizi, K., & Hindarto, D. (2023). Web-based student internship attendance application system for effective student attendance monitoring. *International Journal Software Engineering and Computer Science*. https://journal.lembagakita.org/index.php/ijsecs/article/view/1760

13. Pestijo, J. J. (2023). Phases of work immersion and students performance: Basis for a localized internship program. *International Journal of Social Science Humanity & Management Research*. https://www.ijsshmr.com/v2i9/7.php

14. Sangkudluo, T. (2023). Development of supervision monitoring internship system in educational institutions. *Journal of Library and Information Science Srinakharinwirot University*. https://so02.tci-thaijo.org/index.php/JLIS_SWU/article/view/265901

Prepared by:
De La Cruz, Aaron Karl U.
De Leon, Francine Ann C.
Yamomo, John Kaith L.